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(54) **FIXING STRUCTURE FOR PIPE NUT**

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(57) **ABSTRACT**

A fixing structure for a pipe nut includes a pipe nut main body, a first stopper, and a second stopper. A female screw portion is provided inside of the pipe nut main body, and the pipe nut main body penetrates a vehicle body frame having a hollow cross section. The first stopper is provided at a base end of the pipe nut main body and abuts on an outer surface of the vehicle body frame to restrict movement of the pipe nut main body in a longitudinal direction. The second stopper is provided on the pipe nut main body and abuts on a surface of the vehicle body frame different from the outer surface of the vehicle body frame on which the first stopper abuts, and the second stopper restricts movement of the pipe nut main body in the longitudinal direction.

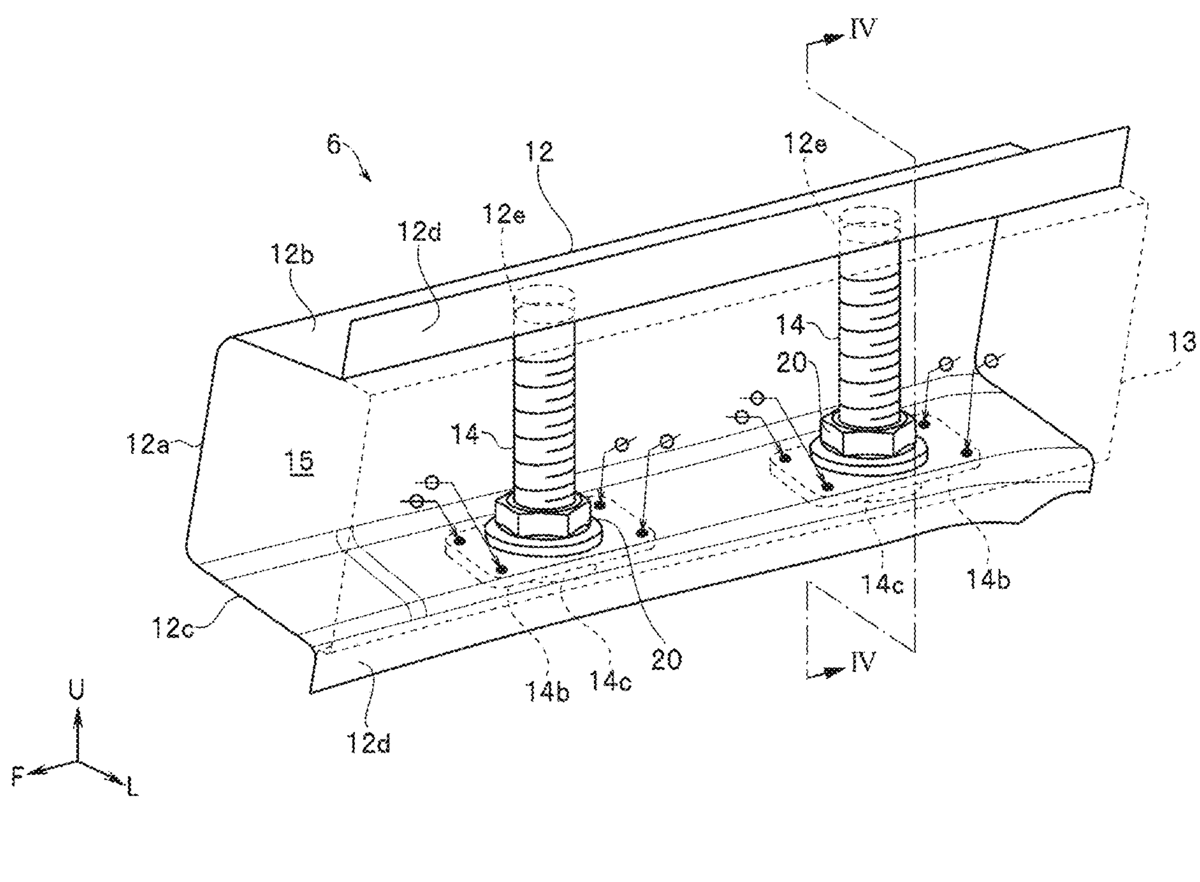
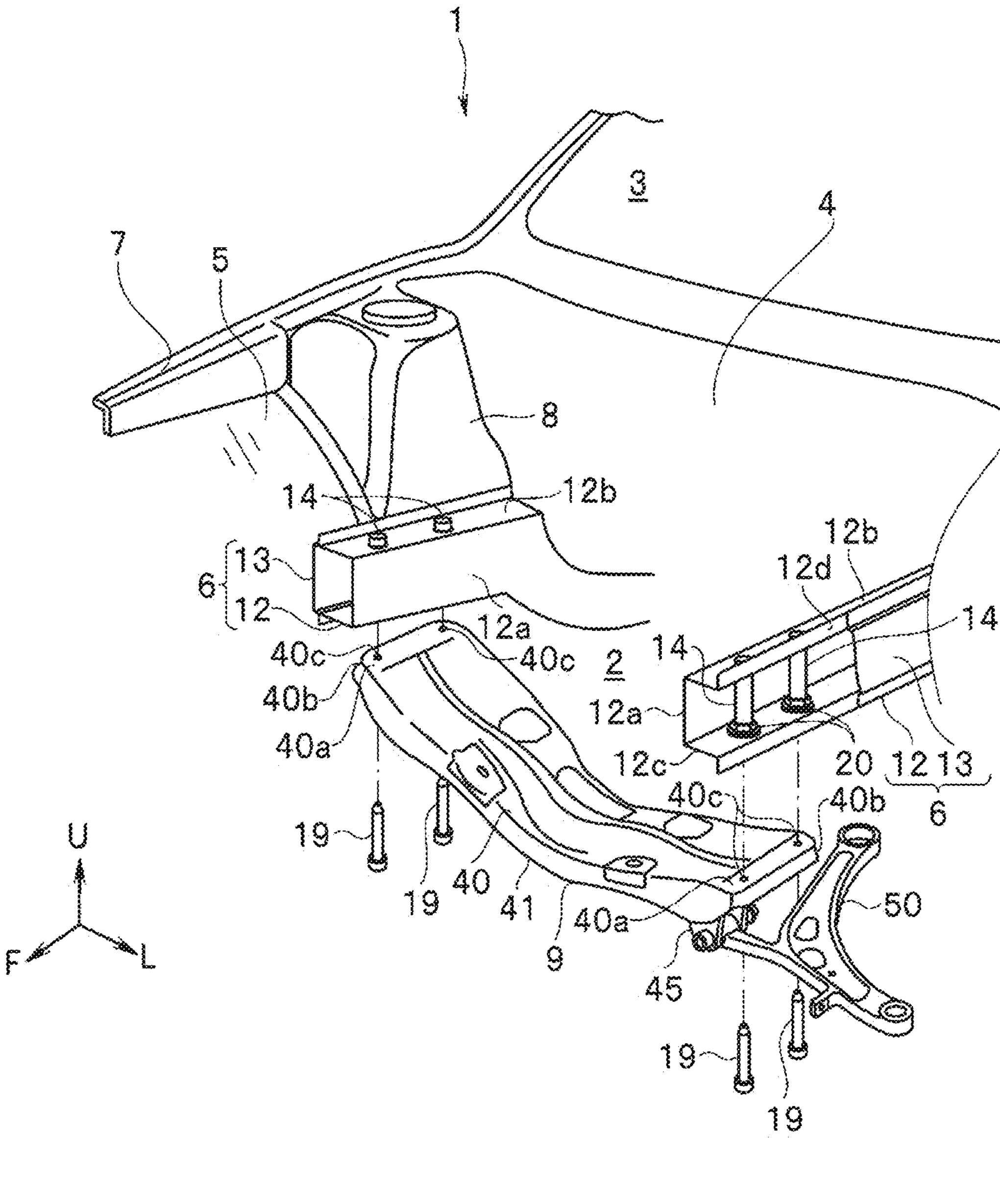


FIG. 1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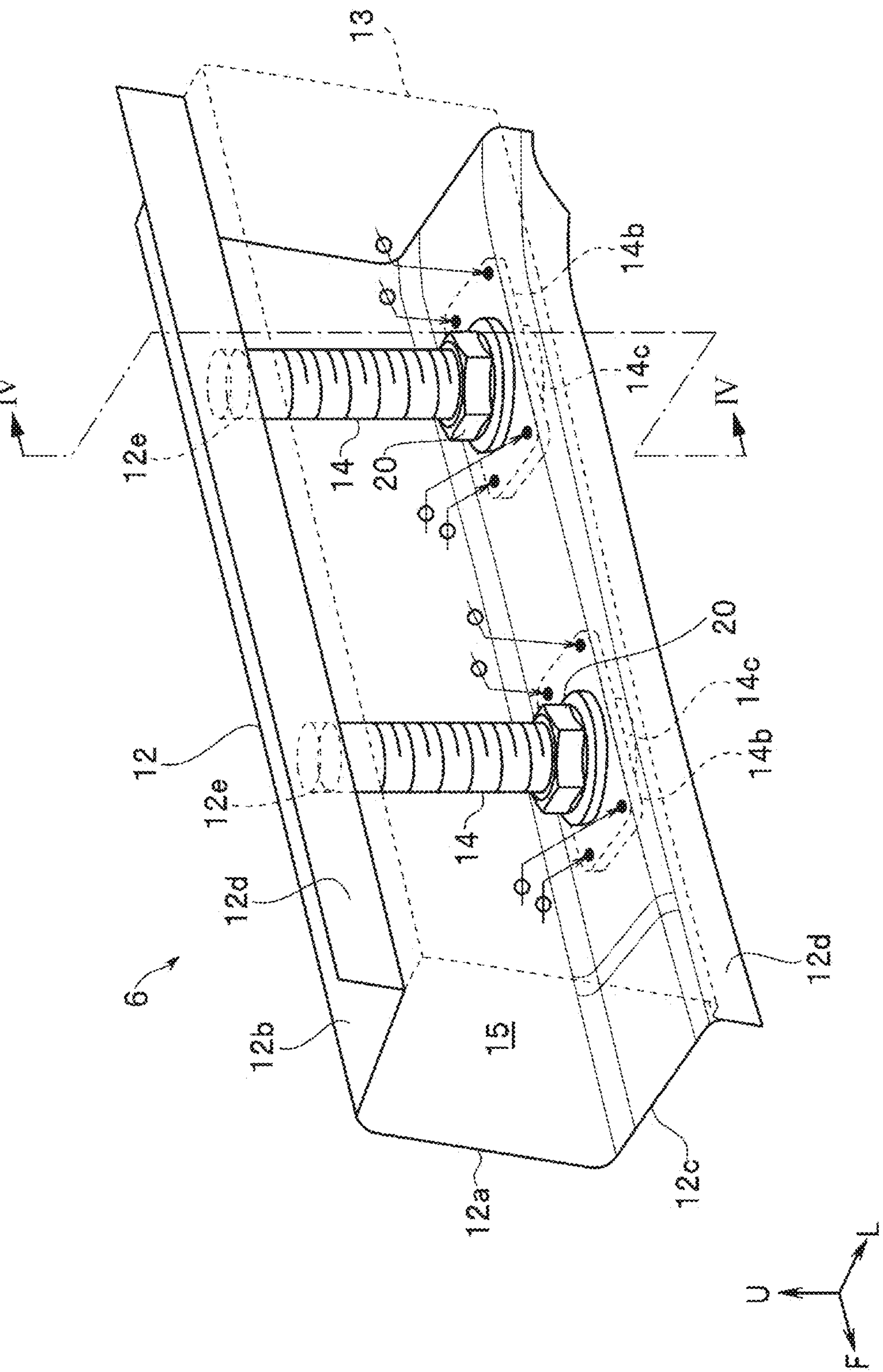
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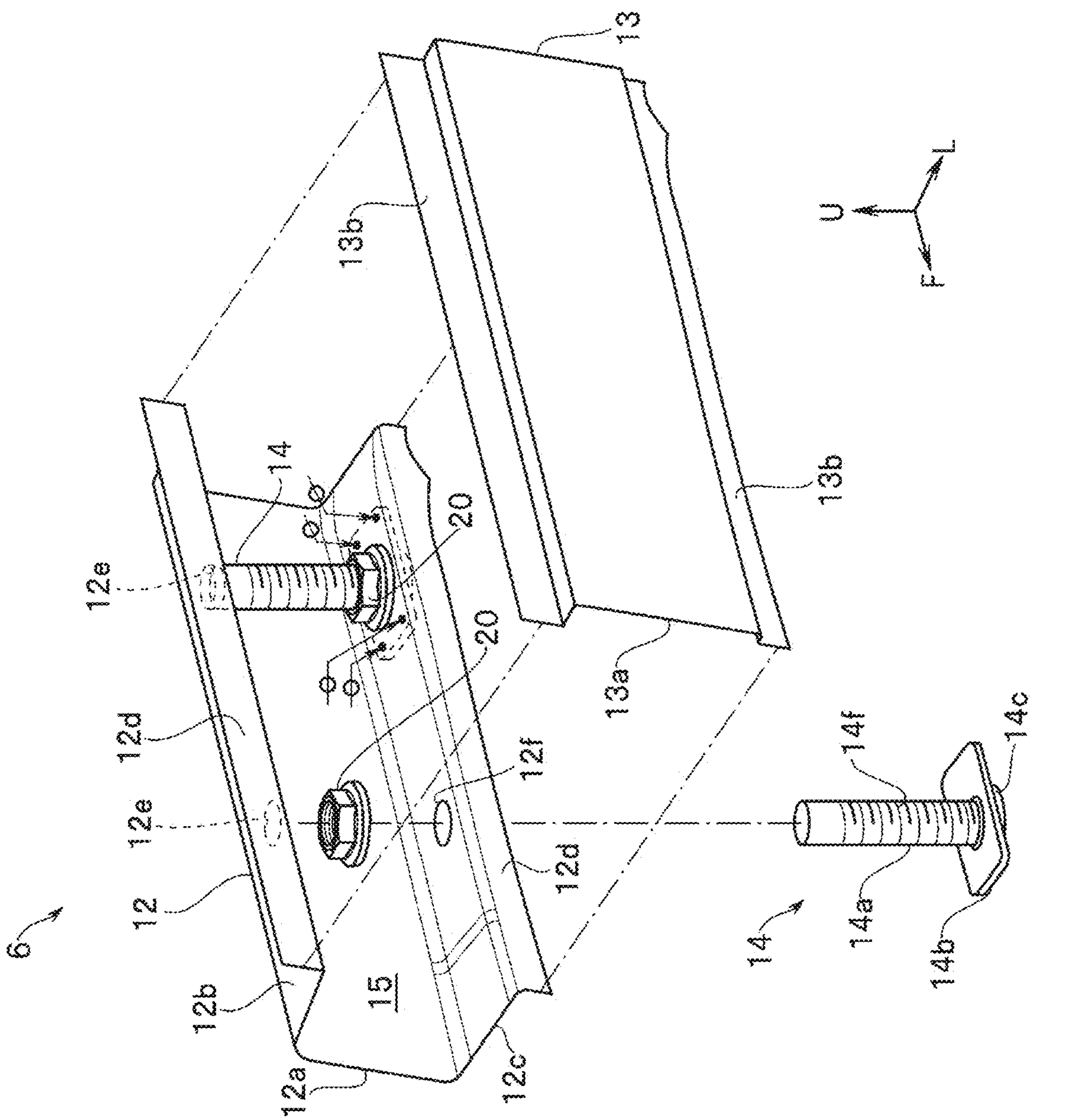
FIG. 2

FIG. 3



6
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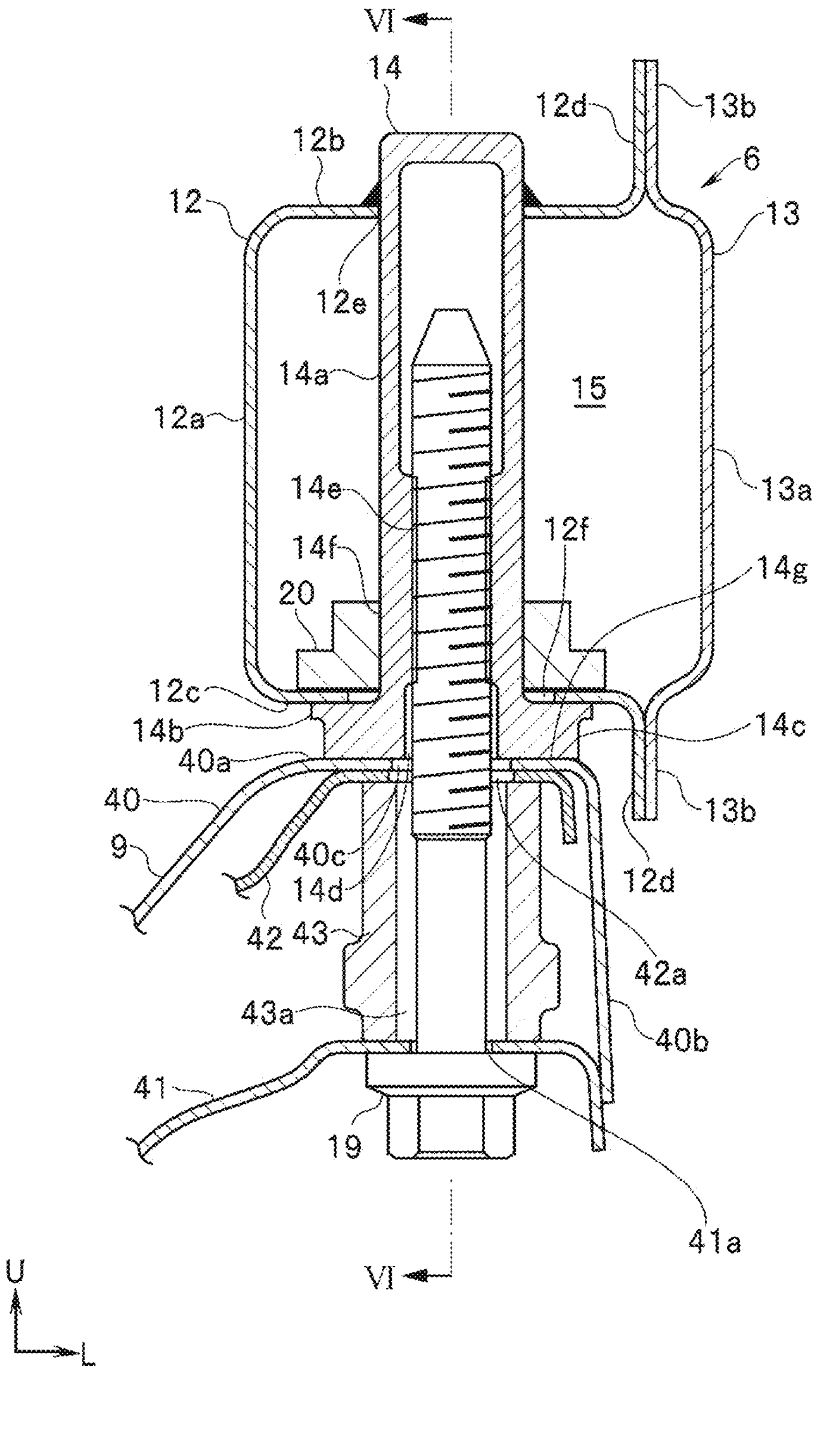
VI
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12c
14b
14c
40a
40
13b
9
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43
42a
43a
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41a
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FIG. 4

FIG. 5

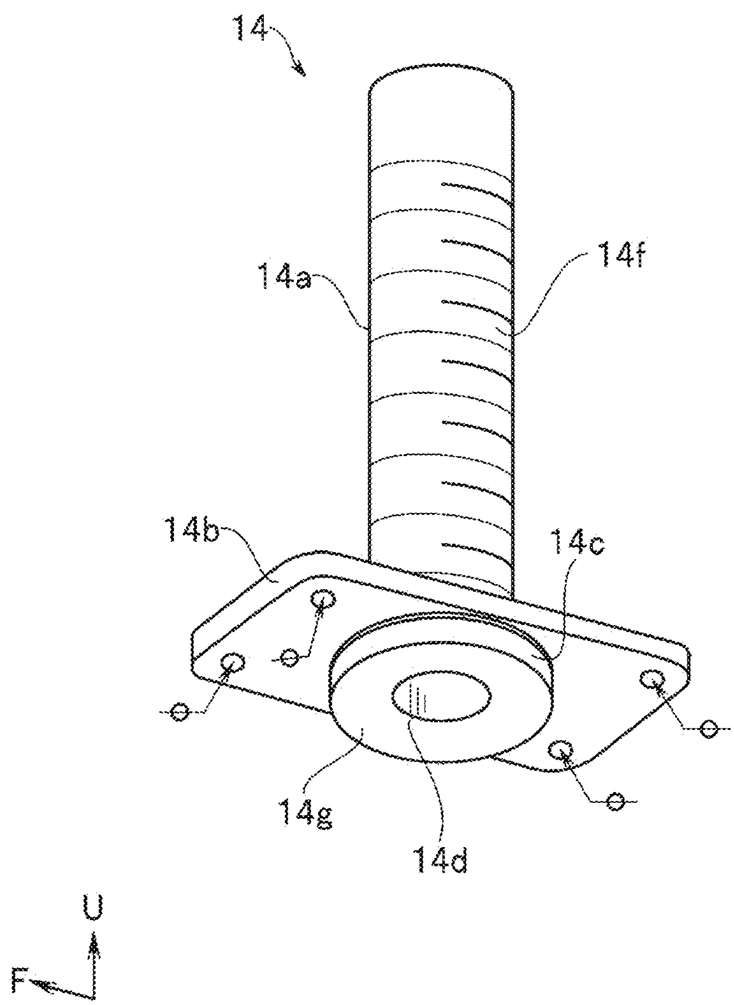

FIG. 6

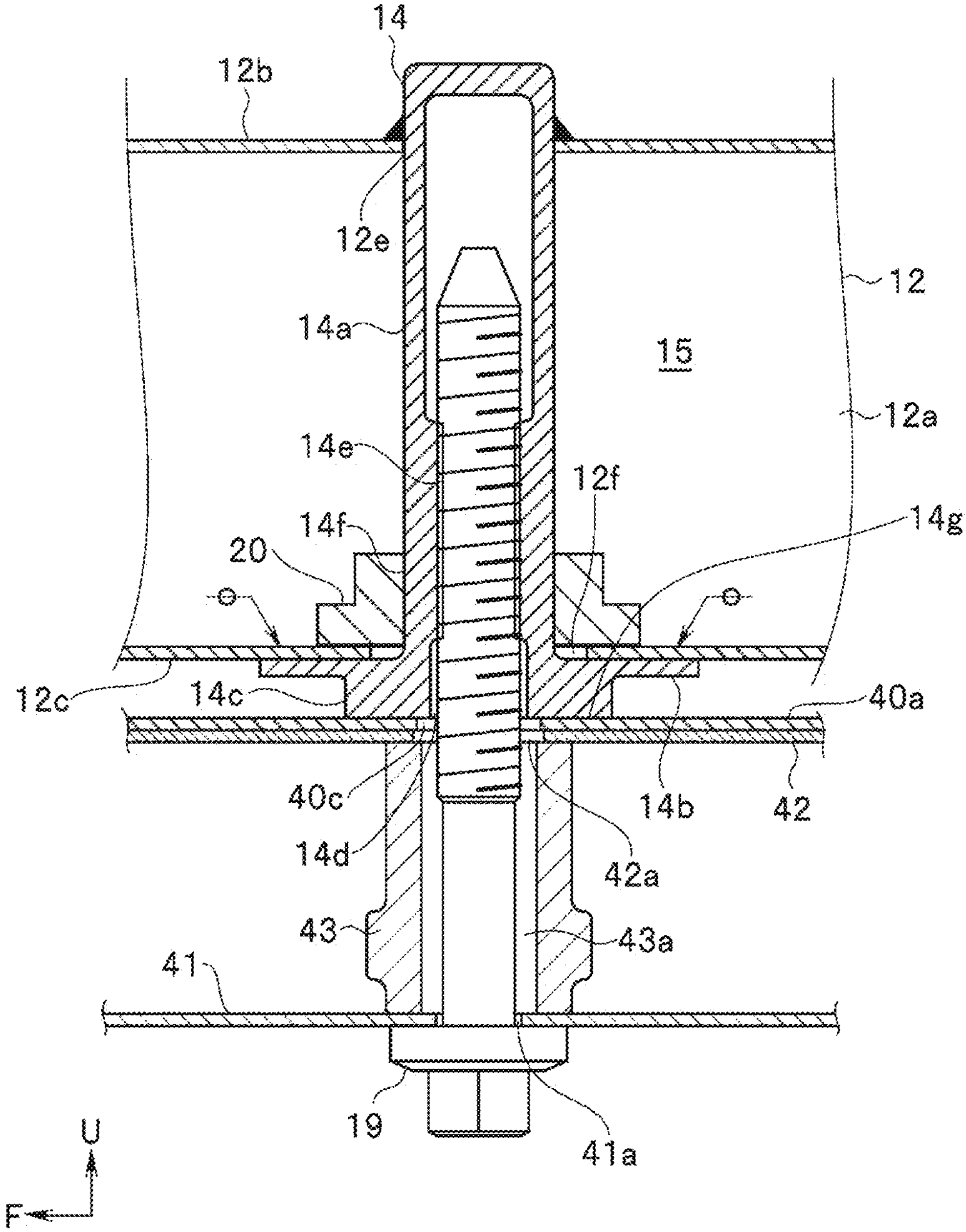
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FIG. 7

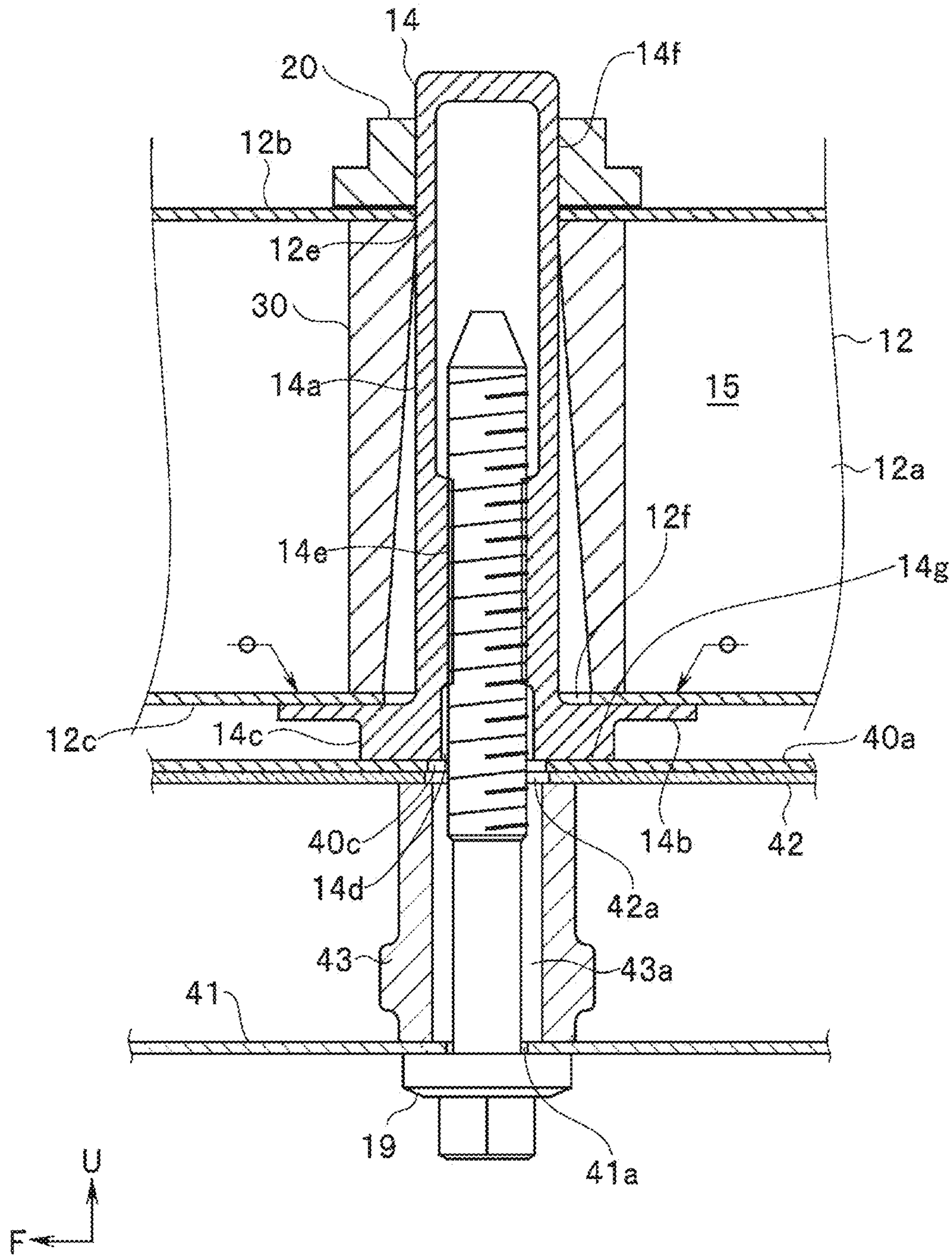
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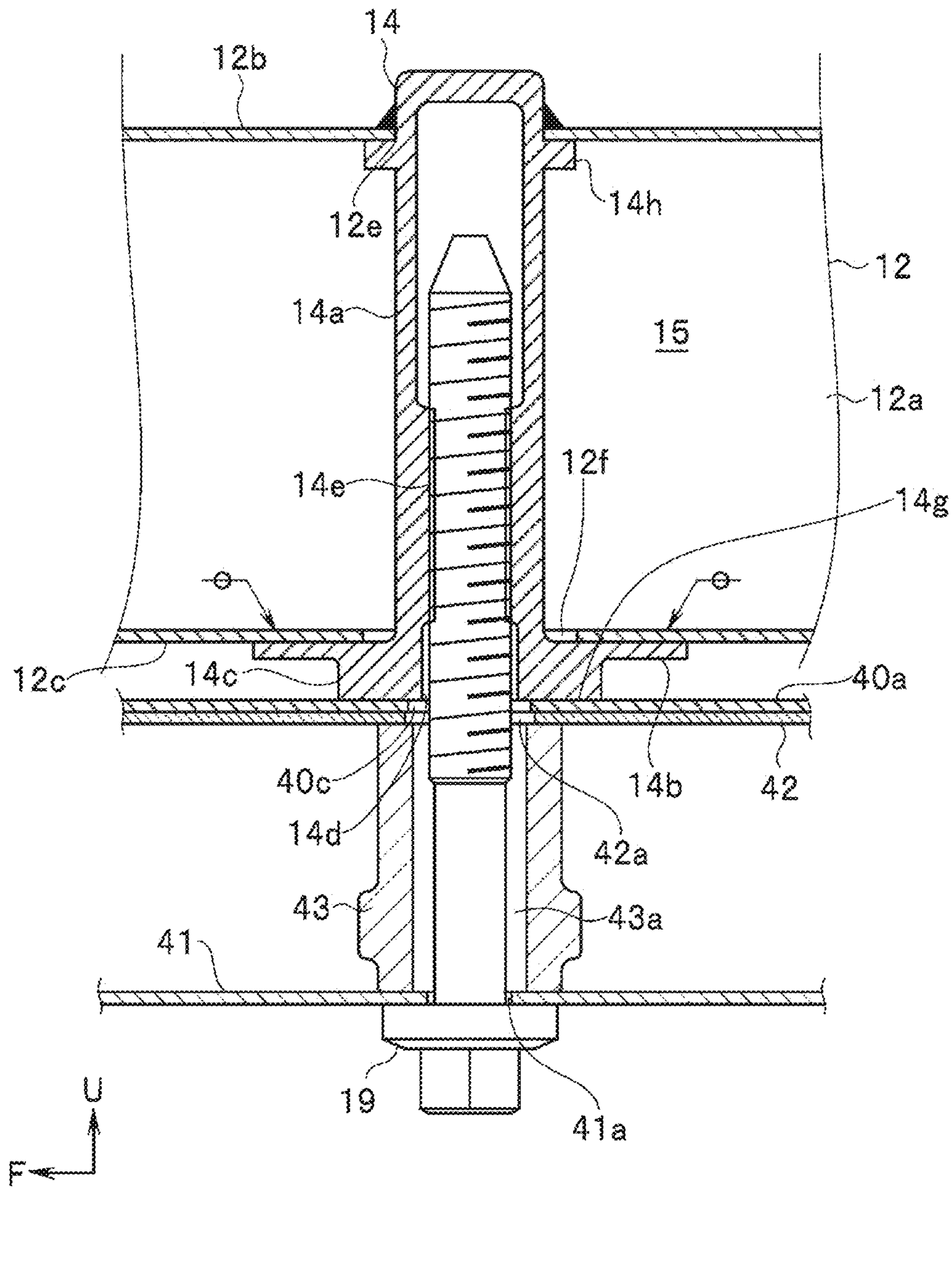
FIG. 8



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FIXING STRUCTURE FOR PIPE NUT

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority from Japanese Patent Application No. 2024-032376 filed on Mar. 4, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

[0002] The disclosure relates to a fixing structure for a pipe nut provided on a vehicle body frame for fastening a sub frame to the vehicle body frame by a bolt.

[0003] Conventionally, front side frames extending in a front-rear direction are disposed as vehicle body frames on both sides of a front portion of a vehicle body of an automobile or the like, for example. On lower portions of the front side frames, for example, a suspension cross member that supports suspensions and the like is extended in the vehicle width direction as a sub frame.

[0004] Both ends of such a suspension cross member are generally attached to each of the front side frames by bolt fastening. Therefore, in order to fasten both ends of the suspension cross member to each of the front side frames by bolts, the front side frames are provided with pipe nuts each having a female screw portion therein.

[0005] The pipe nuts are generally joined to the front side frame by a welding means such as spot welding or arc welding.

[0006] Conventionally, various techniques have been proposed as a fixing structure for a pipe nut for enhancing attachment rigidity.

[0007] For example, in Japanese Patent No. 6337342, a pipe nut protrudes upward in a state of being inserted from below a vehicle body frame. Further, a flange of a proximal end of the pipe nut is brought into contact with a bottom surface of the vehicle body frame. Then, a fixing structure for the pipe nut is disclosed in which an outer periphery of a flange abutting on the bottom surface of the vehicle body frame is joined by arc welding.

[0008] Incidentally, the flange is provided with a bearing surface for abutting an attachment portion of the suspension cross member. In the arc welding for joining the outer periphery of the flange provided with such a bearing surface, spatter occurs during welding. Therefore, during arc welding, it is necessary to protect the bearing surface with a masking material to prevent spatter from adhering. In order to protect the bearing surface with the masking material, a process of attaching and detaching the masking material before and after the arc welding process is separately performed. In the technique disclosed in Japanese Patent No. 6337342, a step of attaching and detaching a masking material is separately performed, and accordingly, there is inconvenience that work man-hours are increased and production efficiency is lowered.

[0009] As a welding means for preventing such a decrease in production efficiency and increasing the production efficiency, for example, welding by spot welding in which spatter hardly occurs is also conceivable.

SUMMARY

[0010] An aspect of the disclosure provides a fixing structure for a pipe nut. The fixing structure for a pipe nut includes a pipe nut main body, a first stopper, and a second stopper. A female screw portion is provided inside of the pipe nut main body, and the pipe nut main body penetrates a vehicle body frame having a hollow cross section. The first stopper is provided at a base end of the pipe nut main body and abuts on an outer surface of the vehicle body frame to restrict movement of the pipe nut main body in a longitudinal direction. The second stopper is provided on the pipe nut main body and abuts on a surface of the vehicle body frame different from the outer surface of the vehicle body frame on which the first stopper abuts, and the second stopper restricts movement of the pipe nut main body in the longitudinal direction.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate an embodiment and, together with the specification, serve to describe the principles of the disclosure.

[0012] FIG. **1** is an exploded perspective view of a vehicle body frame and a suspension cross member according to an embodiment;

[0013] FIG. **2** is an enlarged perspective view of a main part of the vehicle body frame according to the embodiment;

[0014] FIG. **3** is an enlarged exploded perspective view of the main part of the vehicle body frame according to the embodiment;

[0015] FIG. **4** is a cross-sectional view taken along line IV-IV of FIG. **2** according to the embodiment;

[0016] FIG. **5** is a perspective view of a pipe nut according to the embodiment from an obliquely lower side;

[0017] FIG. **6** is a cross-sectional view taken along line VI-VI of FIG. **4** according to the embodiment;

[0018] FIG. **7** is a cross-sectional view corresponding to FIG. **6** according to an embodiment; and

[0019] FIG. **8** is a cross-sectional view corresponding to FIG. **6** according to an embodiment.

DETAILED DESCRIPTION

[0020] In spot welding, it may be difficult to ensure attachment rigidity of a pipe nut as compared with arc welding in which joining is performed by a line.

[0021] It is desirable to provide a fixing structure for a pipe nut capable of enhancing production efficiency and ensuring attachment rigidity of the pipe nut to a vehicle body frame even when the pipe nut is joined to the vehicle body frame by spot welding.

[0022] Hereinafter, embodiments of the disclosure will be described with reference to the drawings.

[0023] Note that, in the drawings used in the following description, scales are different for each component in order to make each component recognizable in the drawings. Therefore, the disclosure is not limited only to the illustrated form with respect to the quantities of components, the shapes of components, the ratios of the sizes of components, the relative positional relationship of components, and the like described in these drawings. Further, in the following description, in a case where "joining" is described, the joining method is performed using a joining means represented by fusion welding, mechanical joining, or the like. Note that, in the drawing, a spot welding symbol is attached to a spot welding point indicating a characteristic portion, and display of the welding symbol is omitted at other welding points.

[0024] As illustrated in FIG. **1**, an engine room **2** is provided in a front portion of the vehicle body **1**.

[0025] The engine room **2** is partitioned from the rear cabin **3** by a toe board **4**.

[0026] As frames constituting such an engine room **2**, the vehicle body **1** includes a pair of left and right front side frames **6**, a pair of left and right upper side frames **7**, and a suspension cross member **9**. These frames are formed by, for example, pressing a sheet metal member made of a high-strength steel plate or the like.

[0027] Note that the basic structure around the pair of left and right front side frames **6** and the pair of left and right upper side frames **7** is bilaterally symmetrical. Therefore, the left side of the vehicle body will be described below as an example.

[0028] As illustrated in FIG. **1**, the front side frames **6** extend from a lower portion of the toe board **4** toward the front of the vehicle body **1** on outsides of the toe board **4** in a vehicle width direction.

[0029] The upper side frames **7** are provided above the front side frames **6** and outside the front side frames **6** in the vehicle width direction. The upper side frames **7** extend forward of the vehicle body along the engine room **2**.

[0030] A suspension tower **8** is provided between each of the front side frames **6** and each of the upper side frames **7** near the rear of the engine room **2**.

[0031] The suspension tower **8** houses a suspension device (not illustrated) therein. The suspension tower **8** supports an upper portion of the suspension device.

[0032] Furthermore, a wheel apron **5** is provided between each of the front side frames **6** and each of the upper side frames **7** in front of the suspension tower **8**.

[0033] The suspension cross member **9** is disposed as a sub frame on lower portions of the front side frames **6**. That is, the suspension cross member **9** extends in the vehicle width direction at the lower portions of the front side frames **6**. The suspension cross member **9** has a shape in which a central portion is curved downward.

[0034] Ends of the suspension cross member **9** in the vehicle width direction are joined to bottom surfaces of the front side frames **6** by bolt fastening.

[0035] In order to fasten such a suspension cross member **9** by bolts, the front side frames **6** as a vehicle body frame are each provided with a pair of front and rear pipe nuts **14**.

[0036] The configurations of the front side frames **6** and the suspension cross member **9** will be described in detail below.

[0037] As illustrated in FIGS. **1** and **2**, the front side frames **6** each include an inner frame **12** and an outer frame **13**. Note that, in FIG. **1**, a part of the outer frame **13** of the left front side frame **6** is omitted.

[0038] As illustrated in FIGS. **2** to **4**, the inner frame **12** has a substantially hat-shaped cross section protruding inward in the vehicle width direction. Such an inner frame **12** includes an inner wall **12***a*, an upper wall **12***b*, a lower wall **12***c*, and a pair of upper and lower inner flanges **12***d*.

[0039] The inner wall **12***a* extends in the vehicle body vertical direction.

[0040] The upper wall **12***b* and the lower wall **12***c* extend outward in the vehicle width direction from both ends of the inner wall **12***a*.

[0041] The pair of upper and lower inner flanges **12***d* extends in the vehicle body vertical direction from outer edges of the upper wall **12***b* and the lower wall **12***c* in the vehicle width direction. Each of the inner flanges **12***d* functions as a flange for joining the inner frame **12** to the outer frame **13**.

[0042] As illustrated in FIGS. **2** to **4**, the outer frame **13** has a substantially hat-shaped cross section with a depth shallower than that of the inner frame **12**. The outer frame **13** is disposed at a position facing the outside of the inner frame **12** in the vehicle width direction. For example, the opening of the outer frame **13** faces the opening of the inner frame **12**.

[0043] The outer frame **13** disposed in this manner has an outer wall **13***a* at a position facing the inner wall **12***a* of the inner frame **12**. The outer frame **13** further includes a pair of upper and lower outer flanges **13***b* at respective positions facing the inner flanges **12***d*.

[0044] Such an outer frame **13** is joined in a state where the outer flanges **13***b* are brought into contact with the inner flanges **12***d*.

[0045] Thus, as illustrated in FIG. **4**, the front side frame **6** has a substantially rectangular cross section having a hollow cross section **15**.

[0046] With respect to the front side frame **6**, the pair of front and rear pipe nuts **14** is provided in a state of penetrating the front side frame **6** in the vertical direction of the vehicle body **1**. Thus, as illustrated in FIGS. **3** and **4**, a pair of front and rear upper insertion holes **12***e* and a pair of front and rear lower insertion holes **12***f* are provided in each of the upper wall **12***b* and the lower wall **12***c* of the front side frame **6**.

[0047] The pair of front and rear upper insertion holes **12***e* and the pair of front and rear lower insertion holes **12***f* are disposed at positions facing each other. That is, the insertion holes **12***e* and **12***f* are arranged substantially coaxially. The diameter of each of the insertion holes **12***e* and **12***f* is set slightly larger than the outer diameter of the pipe nut **14**.

[0048] The pipe nut **14** is inserted into each of the insertion holes **12***e* and **12***f* from below. As illustrated in FIG. **5**, the pipe nut **14** includes a pipe nut main body **14***a*, a joining flange **14***b*, and a boss portion **14***c*.

[0049] As illustrated in FIGS. **4** and **6**, the pipe nut main body **14***a* has, for example, a substantially cylindrical shape with one end closed. The hole formed in the pipe nut main body **14***a* is set as a bolt insertion hole **14***d*. Furthermore, a female screw portion **14***e* is formed on an inner peripheral surface of the bolt insertion hole **14***d*. Note that a bolt **19** to be described later can be screwed into the female screw portion **14***e*.

[0050] A male screw portion **14***f* is formed on an outer peripheral surface of the pipe nut main body **14***a*. The male screw portion **14***f* is formed, for example, from a middle position close to a distal end (upper end) in a longitudinal direction of the pipe nut main body **14***a* to a base end.

[0051] The joining flange **14***b* is formed at the base end of the pipe nut main body **14***a*. As illustrated in FIGS. **2** to **6**, the joining flange **14***b* extends in the front-rear direction of the vehicle body along an outer surface of the lower wall **12***c* of the inner frame **12**. That is, the shape of the joining flange **14***b* is a rectangle that is long in the vehicle body front-rear direction along the lower wall **12***c*.

[0052] Such a joining flange **14***b* functions as a first stopper that restricts the movement of the pipe nut main body **14***a* in the longitudinal direction when abutting on the outer surface of the lower wall **12***c*.

[0053] Further, a portion of the joining flange **14***b* extending in the front-rear direction of the vehicle body functions as a welded portion with the lower wall **12***c*. That is, the joining flange **14***b* allows spot-welding of four corners of the joining flange **14***b* in a state of being in contact with the outer surface of the lower wall **12***c*. By this spot welding, the joining flange **14***b* can be joined to the plate surface of the lower wall **12***c*.

[0054] Here, the length of the joining flange **14***b* in the front-rear direction of the vehicle body is preferably set to be longer than at least the length of the lower wall **12***c* in the vehicle width direction. With such a configuration, when the pipe nut **14** rotates about an axis, the joining flange **14***b* can be brought into contact with the inner flange **12***d*. Thus, the joining flange **14***b* restricts rotation of the pipe nut **14** as a whirl-stop.

[0055] As illustrated in FIG. **5**, the boss portion **14***c* is formed to protrude downward from the joining flange **14***b*. The boss portion **14***c* is formed in an annular shape. The protruding surface of the boss portion **14***c* is provided with a bearing surface **14***g*.

[0056] As illustrated in FIGS. **4** and **6**, an upper surface of the suspension cross member **9** abuts on the bearing surface **14***g*. That is, the bearing surface **14***g* functions as an attachment surface when the suspension cross member **9** is fasten by bolts. Therefore, the bearing surface **14***g* is substantially flat.

[0057] Inside the front side frame **6**, the nut **20** is screwed onto the male screw portion **14***f* of the pipe nut **14** when being inserted into the lower insertion hole **12***f*. Note that the nut **20** is screwed before the outer frame **13** is joined to the inner frame **12**.

[0058] By such screwing onto the male screw portion **14***f*, the nut **20** is disposed at a position in contact with an inner surface of the lower wall **12***c*. Thus, the nut **20** functions as a second stopper that restricts the movement of the pipe nut main body **14***a* in the longitudinal direction.

[0059] Then, the nut **20** sandwiches the plate member of the lower wall **12***c* with the joining flange **14***b*. In this manner, the nut **20** can firmly fix the pipe nut **14** to the lower wall **12***c* of the front side frame **6**.

[0060] Note that the diameter of the nut **20** is set to be larger than the diameter of the lower insertion hole **12***f* in order to accurately sandwich the lower wall **12***c*. In order to achieve such a diameter of the nut **20**, it is preferable to use a flange nut as the nut **20**.

[0061] By the fixing using the joining flange **14***b* and the nut **20**, a state in which the pipe nut main body **14***a* penetrates the lower insertion hole **12***f* of the lower wall **12***c* and also penetrates the upper insertion hole **12***e* of the upper wall **12***b* is maintained.

[0062] As illustrated in FIG. **6**, the joining flange **14***b* of the pipe nut **14** fixed in this manner is joined to the lower wall **12***c* by spot welding. Further, a distal end side of the pipe nut main body **14***a* is joined to an outer surface of the upper wall **12***b* by arc welding in a state of being inserted into the upper insertion hole **12***e*.

[0063] As illustrated in FIGS. **1**, **4**, and **6**, the suspension cross member **9** includes an upper panel **40**, a lower panel **41**, and a reinforcement **42**.

[0064] The upper panel **40** has, for example, an open cross-sectional shape whose cross section is open toward the lower side of the vehicle body. As illustrated in FIG. **4**, such an upper panel **40** includes an attachment portion **40***a* and a side wall **40***b*.

[0065] The attachment portion **40***a* is disposed at a position facing the lower wall **12***c*. The attachment portion **40***a* abuts on the bearing surface **14***g* of the pipe nut **14** and functions as an attachment portion for fixing the suspension cross member **9** to the front side frame **6**.

[0066] Thus, the attachment portion **40***a* has a pair of front and rear first bolt insertion holes **40***c*.

[0067] The pair of front and rear first bolt insertion holes **40***c* is provided at respective positions facing the bolt insertion holes **14***d*. That is, the first bolt insertion holes **40***c* and the bolt insertion holes **14***d* are arranged substantially coaxially.

[0068] The side wall **40***b* extends downward from an outer edge of the attachment portion **40***a* in the vehicle width direction.

[0069] The lower panel **41** is disposed at a position facing the upper panel **40** with a space therebetween. The lower panel **41** is provided with a second bolt insertion hole **41***a* at a position facing each of the first bolt insertion holes **40***c*. That is, the second bolt insertion holes **41***a* and the first bolt insertion holes **40***c* are arranged substantially coaxially.

[0070] The outer edge of the lower panel **41** in the vehicle width direction is bent downward. In this manner, a lower end of the side wall **40***b* of the upper panel **40** is joined to an outer surface of the bent lower panel **41** in the vehicle width direction.

[0071] By joining the upper panel **40** and the lower panel **41**, an internal space is formed in the suspension cross member **9**.

[0072] The reinforcement **42** is disposed inside the suspension cross member **9**. For example, the reinforcement **42** extends in the vehicle width direction along the upper panel **40**.

[0073] As illustrated in FIG. **4**, the outer end of the reinforcement **42** in the vehicle width direction is in contact with the inner surface of the attachment portion **40***a*. Then, an outer end of the reinforcement **42** in the vehicle width direction is joined to the attachment portion **40***a*.

[0074] Further, the joint portion of the reinforcement **42** is provided with a third bolt insertion hole **42***a* at a position facing each first bolt insertion hole **40***c*. That is, each third bolt insertion hole **42***a* and each of the first and second bolt insertion holes **40***c* and **41***a* are arranged substantially coaxially.

[0075] As illustrated in FIGS. **4** and **6**, a pair of front and rear spacers **43** is extended between the reinforcement **42** and the lower panel **41**.

[0076] The pair of front and rear spacers **43** has a substantially cylindrical shape. The hole formed in each spacer **43** is set as a fourth bolt insertion hole **43***a* penetrating in the vehicle body vertical direction. That is, the fourth bolt insertion hole **43***a* of each spacer **43** is disposed substantially coaxially with each of the first to third bolt insertion holes **40***c*, **41***a*, and **42***a*.

[0077] A bolt **19** can be inserted into the first to fourth bolt insertion holes **40***c*, **41***a*, **42***a*, and **43***a* arranged in this manner.

[0078] Note that, in the following description, the first to fourth bolt insertion holes **40***c*, **41***a*, **42***a*, and **43***a* are collectively referred to as a bolt insertion hole H.

[0079] With such a configuration, the suspension cross member **9** is fixed to the front side frame **6** by bolt fastening.

[0080] For example, when the suspension cross member **9** is fixed to the front side frame **6**, the attachment portion **40***a* abuts on each of the bearing surfaces **14***g* from below.

[0081] Further, each bolt insertion hole H is positioned with respect to the bolt insertion hole **14***d* of each of the pipe nuts **14**.

[0082] The bolt **19** is inserted into the bolt insertion hole H and the bolt insertion hole **14***d* of each set positioned from below the bolt insertion hole H. Then, each bolt **19** is screwed into the female screw portion **14***e* of the pipe nut **14**. Thus, the suspension cross member **9** is fixed to the front side frame **6** by fastening.

[0083] An arm support portion **45** is provided on the outside in the vehicle width direction of the suspension cross member **9** fixed in this manner.

[0084] For example, as illustrated in FIG. **1**, the arm support portion **45** protrudes downward from a bottom surface on the outside in the vehicle width direction of the lower panel **41**. A base end of a suspension lower arm **50** is pivotally supported by the arm support portion **45**. Thus, a distal end side of the suspension lower arm **50** is swingable in the vertical direction.

[0085] A lower portion of a suspension device, which is not illustrated, is supported outside the suspension lower arm **50** in the vehicle width direction. Furthermore, a wheel hub, which is not illustrated, for fixing the front wheel is rotatably supported on the outside in the vehicle width direction of the suspension lower arm **50**.

[0086] According to such an embodiment, a fixing structure of the pipe nut **14** includes the pipe nut main body **14***a* in which the female screw portion **14***e* is formed inside, and the pipe nut main body penetrating the front side frame **6** having the hollow cross section **15**, the joining flange **14***b* that is provided at the base end of the pipe nut main body **14***a* and abuts on an outer surface of the front side frame **6** to restrict movement of the pipe nut main body **14***a* in the longitudinal direction, and the nut **20** that is provided on the pipe nut main body **14***a* and abuts on a surface of the front side frame **6** different from the outer surface of the front side frame **6** on which the joining flange **14***b* abuts, the nut **20** restricting movement of the pipe nut main body **14***a* in the longitudinal direction.

[0087] With these configurations, the fixing structure of the pipe nut **14** can enhance production efficiency and ensure attachment rigidity of the pipe nut **14** to the vehicle body frame even when the pipe nut **14** is joined to the vehicle body frame by spot welding.

[0088] That is, in the present embodiment, the pipe nut **14** has the pipe nut main body **14***a* penetrating the front side frame **6**. Further, at the base end of the pipe nut main body **14***a*, the joining flange **14***b* that abuts on the outer surface of the front side frame **6** to restrict movement in the longitudinal direction is provided. Furthermore, the pipe nut main body **14***a* is provided with the nut **20** that abuts on a surface of the front side frame **6** different from the surface on which the joining flange **14***b* abuts and restricts the movement in the longitudinal direction.

[0089] With these configurations, the fixing structure of the pipe nut **14** of the present embodiment can firmly fix the pipe nut **14** to the front side frame **6**. Thus, the pipe nut **14** can generate a sufficient drag against loads in a prying direction (direction inclined with respect to the central axis) and the longitudinal direction input to the pipe nut main body **14***a* via the suspension cross member **9** during traveling. Therefore, even when the joining flange **14***b* is joined to the lower wall **12***c* by spot welding, sufficient attachment rigidity of the pipe nut **14** with respect to the front side frame **6** can be ensured.

[0090] In addition, using spot welding for joining the joining flange **14***b* to the lower wall **12***c* can significantly reduce the amount of spatter (melted metal powder) generated as compared with arc welding. Therefore, the amount of spatter adhering to the bearing surface **14***g* during welding can be significantly reduced. Therefore, in a welding step of the present embodiment in which spot welding can be used, the attaching and detaching step of a masking material for protecting the bearing surface **14***g* is unnecessary. As a result, the fixing structure of the pipe nut **14** of the present embodiment can enhance the production efficiency.

[0091] In addition, the nut **20** sandwiches the plate member of the lower wall **12***c* of the front side frame **6** with the joining flange **14***b* inside the front side frame **6**. That is, the joining flange **14***b* and the lower wall **12***c* are brought into pressure contact with each other by the nut **20**. Thus, when the joining flange **14***b* is spot-welded to the lower wall **12***c*, welding can be performed in a state where the contact area between the joining flange **14***b* and the lower wall **12***c* is enlarged. Therefore, welding accuracy at the time of spot welding can be enhanced.

[0092] Further, the distal end of the pipe nut main body **14***a* protrudes from the upper wall **12***b* in a state of being inserted into the upper insertion hole **12***e* of the front side frame **6**. That is, the distal end of the pipe nut main body **14***a* protrudes to a position above and away from the bearing surface **14***g*. Thus, the amount of spatter generated when the pipe nut main body **14***a* is welded to the upper wall **12***b* reaches the bearing surface **14***g* is limited. Therefore, even when the distal end of the pipe nut main body **14***a* is firmly joined by arc welding, a masking material for protecting the bearing surface **14***g* is unnecessary. As a result, in the fixing structure of the pipe nut **14** of the present embodiment, the distal end of the pipe nut **14** can be fixed by arc welding without lowering the production efficiency.

[0093] Further, the joining flange **14***b* has a rectangular shape elongated in the vehicle body front-rear direction along the lower wall **12***c*. Thus, when the joining flange **14***b* abuts on the outer surface of the lower wall **12***c*, it is possible to accurately restrict the movement of the pipe nut main body **14***a* in the longitudinal direction.

[0094] Further, by extending in the front-rear direction of the vehicle body, sufficient welding regions for spot welding can be ensured at the four corners of the joining flange **14***b* sufficiently separated from the bearing surface **14***g*.

[0095] Further, the length of the joining flange **14***b* in the front-rear direction of the vehicle body is set to be longer than at least the length of the lower wall **12***c* in the vehicle width direction. That is, the joining flange **14***b* has a length capable of abutting on the inner flange **12***d*. Therefore, rotation of the pipe nut **14** can be restricted without adding a new structure to the joining flange **14***b*. Then, by restricting the rotation of the pipe nut **14** in this manner, it is possible to improve workability when the pipe nut **14** is assembled to the front side frame **6**.

[0096] Note that, although the setting of the shape and the like of the joining flange **14***b* has been described as an example of the whirl-stop of the pipe nut **14**, a notch may be provided in a part of the distal end of the pipe nut main body **14***a* to restrict the rotation of the pipe nut **14**. In this case, the shape of the upper insertion hole **12***e* is changed to a shape corresponding to the notch of the distal end of the pipe nut main body **14***a*.

[0097] FIG. **7** illustrates a second embodiment of the disclosure. Components similar to those in the first embodiment are denoted by similar reference numerals, and description thereof is simplified or omitted.

[0098] The nut **20** of the first embodiment sandwiches the lower wall **12***c* of the front side frame **6** with the joining flange **14***b*.

[0099] On the other hand, a nut **20** of the present embodiment sandwiches the front side frame **6** with the joining flange **14***b*. Note that, in the present embodiment, the male screw portion **14***f* is formed, for example, from a distal end to a central portion in the longitudinal direction of the pipe nut main body **14***a*.

[0100] For example, the nut **20** is screwed onto the male screw portion **14***f* outside the front side frame **6** when the male screw portion **14***f* is inserted into the upper insertion hole **12***e*.

[0101] By such screwing with the male screw portion **14***f*, the nut **20** is disposed at a position in contact with the outer surface of the upper wall **12***b*. Thus, the nut **20** restricts the movement of the pipe nut main body **14***a* in the longitudinal direction.

[0102] Then, the nut **20** sandwiches the front side frame **6** with the joining flange **14***b*.

[0103] The front side frame **6** sandwiched in this manner has a spacer **30** therein.

[0104] The spacer **30** is disposed so as to cover the outer periphery of the pipe nut main body **14***a*. That is, the spacer **30** has a substantially cylindrical shape through which the pipe nut main body **14***a* can be inserted.

[0105] Both ends of the spacer **30** in the vehicle body vertical direction are fixed in a state of being in contact with an inner surface of the front side frame **6**. Note that at least one of both ends of the spacer **30** is preferably joined to the inner surface of the front side frame **6**.

[0106] With such a configuration, the fixing structure of the present embodiment can firmly fix the pipe nut **20** to the front side frame **6** just by screwing the nut **14** onto the distal end of the pipe nut main body **14***a*.

[0107] That is, the nut **20** in the present embodiment also functions as a joining means for fixing the distal end of the pipe nut main body **14***a* to the upper wall **12***b*. This makes it possible to omit the welding step by arc welding as in the first embodiment. Therefore, the fixing structure of the pipe nut **14** of the present embodiment can enhance the production efficiency as compared with the first embodiment.

[0108] In addition, the front side frame **6** includes the spacer **30** therein. Such a spacer **30** generates a drag against a pressing force concentrated on the outer surface of the front side frame **6** when the nut **20** is screwed onto the distal end of the pipe nut main body **14***a*. Thus, even if the nut **20** is tightened with a strong force, deformation of the plate material (deformation of the upper wall **12***b* and the lower wall **12***c*) of the front side frame **6** can be prevented. Accordingly, tightening strength of the nut **20** with respect to the front side frame **6** can be increased. Therefore, even in a structure in which the pipe nut **14** is fixed by the nut **20** from the outside of the front side frame **6**, sufficient attachment rigidity of the pipe nut **14** with respect to the front side frame **6** can be ensured.

[0109] Further, in the case of the fixing structure of the present embodiment, it is also possible to improve the service performance by omitting the spot welding.

[0110] That is, by omitting the spot welding at the time of joining the joining flange **14***b* to the lower wall **12***c*, the pipe nut **14** can be easily replaced just by releasing the screwing of the nut **20**. Thus, the pipe nut **14** can be set as a replacement part depending on the vehicle type to be applied. Therefore, in the case of the fixing structure of the present embodiment, service performance can be improved by omitting the spot welding.

[0111] Since the other configurations are similar to those of the first embodiment, the description thereof will be omitted.

[0112] FIG. **8** illustrates a third embodiment of the disclosure. Components similar to those in the first and second embodiments are denoted by similar reference numerals, and description thereof is simplified or omitted.

[0113] In the present embodiment, as a second stopper, the positioning flange **14***h* is integrally formed at an intermediate position on the distal end side of the pipe nut main body **14***a*.

[0114] For example, as illustrated in FIG. **8**, the positioning flange **14***h* is annularly projected on the outer periphery on the distal end side of the pipe nut main body **14***a*. That is, the outer diameter of the positioning flange **14***h* is set to be larger than the diameter of the upper insertion hole **12***e*.

[0115] With such a setting, when the distal end of the pipe nut main body **14***a* is inserted into the upper insertion hole **12***e*, the upper surface of the positioning flange **14***h* is positioned in a state of being in contact with the inner surface of the upper wall **12***b*.

[0116] Further, when the positioning flange **14***h* comes into contact with the upper wall **12***b*, the joining flange **14***b* is in contact with the outer surface of the lower wall **12***c*.

[0117] Then, the distal end of the positioned pipe nut main body **14***a* is joined to the upper wall **12***b* by arc welding. Furthermore, the joining flange **14***b* abutting on the outer surface of the lower wall **12***c* is joined by spot welding.

[0118] Note that, in the present embodiment, the diameter of the lower insertion hole **12***f* is set to be slightly larger than the outer diameter of the positioning flange **14***h*. Accordingly, when the distal end of the pipe nut main body **14***a* is inserted into the lower insertion hole **12***f*, the positioning flange **14***h* is inserted without interfering with the lower wall **12***c*.

[0119] With a simpler configuration than the first embodiment, such a pipe nut **14** can generate a drag against a load in the prying direction and the longitudinal direction input to the pipe nut main body **14***a* via the suspension cross member **9** during traveling. That is, the configuration can be simplified by employing the fixing structure using the pipe nut **14** of the present embodiment according to the vehicle type.

[0120] Since the other configurations are similar to those of the first embodiment, the description thereof will be omitted.

[0121] Note that the disclosure described in the above embodiments (first to third embodiments) is not limited to those embodiments, and various modifications can be made in the implementation stage without departing from the gist thereof.

[0122] For example, the fixing structure of the pipe nut **14** described in each of the above embodiments (first to third embodiments) can also be used for a fastening portion between a vehicle body and a seat, a fastening portion between a vehicle body and a battery case mounted on an electric vehicle, or the like.

[0123] Furthermore, the above-described embodiments include inventions at various stages, and various inventions can be extracted by appropriately combining disclosed constituent elements.

[0124] Further, for example, when the problem to be solved in the disclosure can be solved and the described effect can be obtained even if some constituent elements are deleted from all the constituent elements described in the above embodiments, the configuration from which the constituent elements are deleted can be extracted as an invention.

1. A fixing structure for a pipe nut, the fixing structure comprising:

- a pipe nut main body inside of which a female screw portion is provided, the pipe nut main body penetrating a vehicle body frame having a hollow cross section;
- a first stopper that is provided at a base end of the pipe nut main body and abuts on an outer surface of the vehicle body frame to restrict movement of the pipe nut main body in a longitudinal direction; and
- a second stopper that is provided on the pipe nut main body and abuts on a surface of the vehicle body frame different from the outer surface of the vehicle body frame on which the first stopper abuts, the second stopper restricting movement of the pipe nut main body in the longitudinal direction.

2. The fixing structure for a pipe nut according to claim **1**, wherein

- the second stopper is a nut screwed onto an outer periphery of the pipe nut main body, and
- the second stopper sandwiches a plate member of the vehicle body frame on which the first stopper abuts with the first stopper.

3. The fixing structure for a pipe nut according to claim **1**, wherein

- the second stopper is a nut screwed onto an outer periphery of the pipe nut main body, and
- the second stopper sandwiches the vehicle body frame with the first stopper.

4. The fixing structure for a pipe nut according to claim **3**, further comprising a spacer that is provided on an outer periphery of the pipe nut main body and has both ends in contact with an inner surface of the vehicle body frame inside the vehicle body frame.

5. The fixing structure for a pipe nut according to claim **1**, wherein the first stopper is joined to an outer surface of the vehicle body frame by spot welding.

6. The fixing structure for a pipe nut according to claim **2**, wherein the first stopper is joined to an outer surface of the vehicle body frame by spot welding.

7. The fixing structure for a pipe nut according to claim **3**, wherein the first stopper is joined to an outer surface of the vehicle body frame by spot welding.

8. The fixing structure for a pipe nut according to claim **4**, wherein the first stopper is joined to an outer surface of the vehicle body frame by spot welding.

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